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MILITARY SPECIFICATION

DUMMY LOAD - WATTMETER; DA-43/U

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 Scope.- The equipment covered by this specification shall be designed to provide a power meter and a dummy load for use in simulating the antenna of certain types of aircraft radio equipment under test, particularly the AN/ARC-2 and AN/ART-13.

1.2 Classification.- The equipment covered by this specification shall be of one type and shall consist of the following:

Major Unit	Type Designation	Max Over-All Dimensions	Max Allowable Weight lb
Dummy Load - Wattmeter	DA-43/U	8-1/2 x 9-1/2 x 12 in.	25

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Military

MIL-C-5015	Connectors; Electrical
MIL-T-945	Test-Equipment, for Use with Electronic Equipment; General Specification
JAN-P-658	Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical, and Electro-Mechanical)

Bureau of Aeronautics

SR-197	Nomenclature and Nameplates for Electronic and Associated Equipment

2.1.2 Other Publications

Military Standard (Book)

MIL-STD-129	Marking of Shipments

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 The materials, component parts, mechanical assemblies, and workmanship used in the construction of this equipment shall be in accordance with Specification MIL-T-945.

3.2 General Design Requirements.-

3.2.1 General.- This equipment shall meet all the general design requirements of Specification MIL-T-945.

3.2.2 Total Weight.- The total weight of the equipment, including accessories stowed within the lid of the combination case, shall not exceed 25 pounds.

3.2.3 Case.- The equipment shall be furnished in a combination case of aluminum in accordance with paragraph 3.51.2.2 of Specification MIL-T-945.

3.2.3.1 Size.- The over-all dimensions of the combination case, including cover, handles, hinges, and clasps shall not exceed 8-1/2 x 9-1/2 x 12 inches.

3.2.4 Safety.- The equipment shall contain suitable safety features to prevent indicator failure by accidental connection or switching of high-power transmitters to low-powered input circuits.

3.2.5 Blower.- A 28-volt d-c operated blower shall be provided to circulate air through and around the dummy-load resistors. The blower shall be thermostatically controlled in such a manner that the temperature inside the case will not exceed a reasonably safe value.

3.2.5.1 Filtering.- A suitable filtering network shall be provided in the blower motor circuit to suppress objectionable radio interference.

3.2.6 Component Spacing.- The wiring and component spacing within the case shall be such as to withstand voltages of 10 kilovolts without breakdown.

3.2.7 Equipment Accessories.- The following accessories shall be furnished with each equipment.

3.2.7.1 Power Cable.- The blower motor power cable shall be a 15-foot length of multiconductor cable, terminated at each end with a connector capable of mating with a type AN3102-16-11S receptacle. The cable shall be securely stowed inside the equipment case cover.

3.2.7.2 Tools.- Allen wrenches or other suitable tools necessary for adjusting the front panel switches shall be provided with the equipment and shall be securely fastened inside the case cover.

3.2.8 Identification of Product.-

3.2.8.1 Nameplate.- The equipment nameplate shall be in accordance with Specification SR-197.

3.2.9 Performance.-

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3.2.9.1 Operating Power.- The performance specified herein shall be obtained without the use of any power source, other than the power required to operate the blower motor. Specifications covering d-c power are set forth in paragraph 3.48.2.2 of Specification MIL-T-945.

3.2.9.2 Service Conditions.- The performance specified herein shall be obtained under any combination of service conditions, as specified in Specification MIL-T-945.

3.2.9.3 Frequency.- The equipment shall be capable of performing the functions stated herein throughout the frequency range from 200 kilocycles to 20 megacycles.

3.2.9.4 Power Absorption.- The power absorption capabilities of the equipment when operating under the conditions specified herein shall be from 2 to 100 watts continuous wave.

3.3 Detailed Design Requirements.-

3.3.1 Equipment Composition.- The equipment shall consist of the following:

(a) Dummy Load

(b) Power Meter

3.3.2 Dummy-Load Characteristics.- The dummy load shall consist essentially of a capacitance in series with a resistance, with provision for independent selection of five capacitance values and five resistance values. The voltage drop across the resistance element shall be the basis for measurement of power.

3.3.2.1 Power Resistors.- The power resistors shall be highly stable, noninductive, and shall be combined in a suitable arrangement to permit a maximum power of 120 watts to be dissipated in any switch position. The resistance values shall be such as to permit a front panel selection of either 4, 6, 10, 20, or 40 ohms.

3.3.2.2 Dummy-Load Capacitors.- The four dummy-load capacitors shall be 100 micro-microfarad units with a working d-c rating of 15 kilovolts. Their physical size and arrangement shall be such as to reduce to a negligible value any parallel resonance below 20 megacycles due to stray capacitance. Front panel selection shall make available either 0, 100, 200, 300, or 400 micro-microfarads.

3.3.2.3 Selector Switch.- A dual switch shall be used in order to provide selection of both resistance and capacitance from the front panel. The switch shall be rated at 5 amperes rf, with the capacitor portion sufficiently insulated to withstand 10 kilovolts to ground.

3.3.3 Power-Meter Characteristics.- The power-meter circuit shall be connected to the output of the dummy load and shall consist of the following:

(a) Voltage divider

(b) Crystal rectifier circuit

(c) Meter circuit switch

(d) Power indicator

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3.3.3.1 Voltage Divider Circuit.- A voltage divider circuit shall be incorporated in the power meter to reduce the voltage applied to the crystal diodes. The function of this circuit is to prevent excessive currents through the crystals, or excessive voltage applied to the crystals at any setting of the dummy-load resistance switch.

3.3.3.2 Crystal Rectifier Circuit.- A suitable compensating network shall be provided on the r-f side of each crystal rectifier to minimize the effect of stray capacitance and inductance, and maintain constant voltage on the crystals with change in frequency.

3.3.3.3 Meter Circuit Switch.- A meter circuit switch used for selecting the proper crystal channel shall be attached to the resistance part of the dual switch outlined in paragraph 3.3.2.3 of this specification. The meter switch shall be activated by the same shaft as the resistance switch of the dummy load, with the rotor of the resistance switch being insulated from the shaft in order to isolate rf from the rotor of the meter switch.

3.3.3.4 Power Indicator.- The power indicator shall be a compensated 0 to 1 milliampere meter movement with a nonlinear scale calibrated in watts. The power meter scale shall be from 0 to 100 watts.

3.3.3.5 Allowable Error.- The error of power indication within the range of 2 to 100 watts shall not exceed 10 percent, or 1 watt, whichever is greater.

3.3.4 Front Panel Controls.- The following controls, labeled as indicated shall be provided on the front panel.

3.3.4.1 "SERIES CAP. MMF.".- This shall be a five-position rotary switch and shall indicate the 100, 200, 300, 400 mmf, and "OUT" positions.

3.3.4.2 "SERIES RES. OHMS".- This shall be a five-position rotary switch and shall indicate the 4, 6, 10, 20, and 40 ohm positions.

3.3.4.3 "BLOWER".- This shall be a two-position toggle switch used to apply 28-volt d-c power to the blower motor. This switch shall be suitably labeled for the "ON" and "OFF" positions.

3.3.5 Front Panel Accessories.- The following accessories, labeled as indicated, shall be provided on the front panel.

3.3.5.1 "GROUND".- A binding post shall be provided suitable for making external ground connections.

3.3.5.2 Fuses.- Two fuse containers shall be provided, one having a 0.5 ampere fuse connected in the 28-volt power line and labeled "0.5 AMP". The other container shall hold a spare 0.5 ampere fuse and shall be labeled "SPARE".

3.3.5.3 Indicator Lamp.- A power indicator lamp shall be provided in accordance with paragraph 3.45 of Specification MIL-T-945.

3.3.5.4 Input Power Receptacle.- A type AN3102-16-11S receptacle shall be furnished for providing 28-volt d-c power to the blower motor.

3.3.5.5 Input Insulator.- A suitable input lead-through insulator shall be provided to receive the output of the equipment under test.

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4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of Tests.- The inspection and testing of equipment shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples which are representative of the production of the item after the award of contract, to determine that the production meets the requirements of this specification.

(b) Inspection tests: Inspection tests are those tests accomplished on equipment manufactured and submitted for acceptance under contract.

4.2 Preproduction Tests.-

4.2.1 Sampling Instructions.- Preproduction tests shall be conducted on three preproduction models representative of the production equipment. Additional parts, materials, and design data shall be submitted in accordance with paragraph 4.2.1.2 of Specification MIL-T-945. Samples shall be identified as required and forwarded to the laboratory designated by the procuring agency.

4.2.2 Tests.- Preproduction tests of dummy load - wattmeters shall consist of all the tests required by paragraph 4.2 of Specification MIL-T-945. In addition, the procuring agency reserves the right to make any additional tests deemed necessary to insure compliance with this specification and the contract.

4.2.3 Presubmission Testing.- Prior to the submission of any models for preproduction testing, the contractor shall furnish facilities for and shall conduct tests as required in paragraph 4.2.4 of Specification MIL-T-945. In addition, the data required in paragraph 4.2.3 of Specification MIL-T-945 shall be submitted to the procuring agency for review and approval prior to submission of any models.

4.2.4 Model Acceptance.- Acceptance of the preproduction model shall be by the procuring agency upon satisfactory completion of all tests. No additional models shall be delivered prior to the approval of the preproduction model. Prefabrication of any equipments prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of the model to be submitted for acceptance under the contract. The preproduction model shall not be considered as one of the models under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production model.

4.3 Inspection Tests.- The contractor shall furnish all models and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.3.1 Tests.- Inspection tests shall be in accordance with Specification MIL-T-945, and shall be of the following two types.

4.3.2 Individual Tests.- Individual tests shall be conducted on each model submitted for acceptance under the contract.

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4.3.3 Sampling Tests.- Sampling tests shall be conducted on a quantity as approved by the procuring agency. Models submitted for Sampling test shall be selected by the Government Inspector and shall first have passed the Individual tests. Sampling tests shall be more extensive than the Individual tests and may include any of the tests listed under Preproduction tests which are deemed necessary by the Inspector to determine that production models continue to be equivalent in performance and construction of the approved preproduction model.

4.4 Test Procedures.- The procedures for conducting presubmission testing of preproduction models and inspection testing of production models shall be determined by the contractor. These test procedures shall be subject to review and approval by the procuring agency prior to the initiation of any testing.

4.5 Acceptance of Production Models.- Acceptance of all production models shall be by the cognizant Government Inspector.

5. PREPARATION FOR DELIVERY

5.1 General.- Unless otherwise specified, the equipment and spare parts shall be preserved, packaged, and packed in accordance with the requirements of Specification JAN-P-658. Suitable interior packing or blocking and bracing shall be used to prevent shifting in the shipping container and shock from handling in transit.

5.2 Marking and Labeling.- In addition to any special marking required by the contract or order, marking of interior packages and shipping containers shall be in accordance with the requirements of the individual Services, as specified in MIL-STD-129.

6. NOTES

6.1 Ordering Data.- Invitation for bids and contracts or order shall specify the title, number, and date of this specification.

6.2 Precedence.- When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

6.2.1 Contract.- The contract shall have precedence over any specification.

6.2.2 This Specification.- This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, unless modified by the contract or from subsidiary specifications, unless modified by this specification or the contract, shall be specifically approved in writing by the procuring agency.

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